

4 Rheumatoid arthritis is an inflammatory condition caused by the overactivity of the immune system.

- (a) The photographs show the hand of a person with rheumatoid arthritis and the hand of a person without rheumatoid arthritis.



(Source: © Mediscan/Alamy Stock Photo)

Hand with rheumatoid arthritis

Magnification $\times 0.2$



Hand without rheumatoid arthritis

Magnification $\times 0.2$

- (i) Give **one** piece of evidence, shown in the photographs, that rheumatoid arthritis is an inflammatory condition.

(1)

- (ii) Give **two other** signs of inflammation that are experienced in the hand of a person with rheumatoid arthritis.

(1)

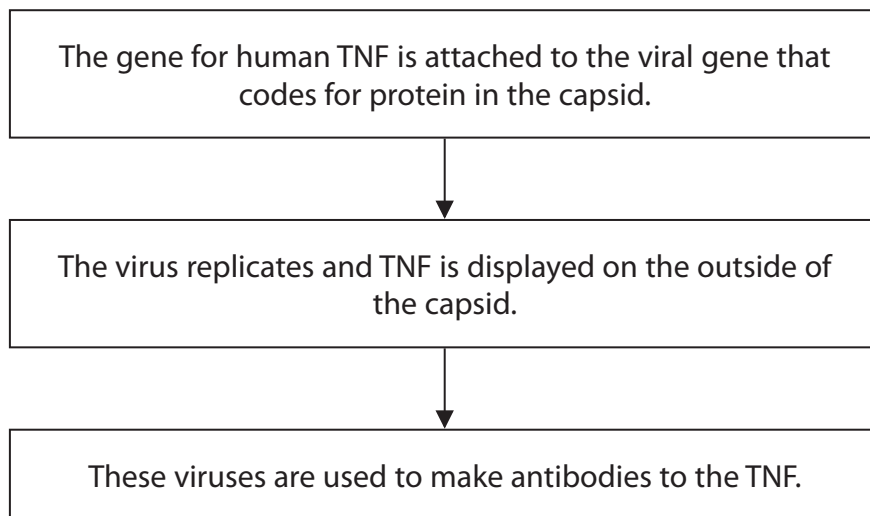
- (b) Tumour necrosis factor (TNF) is a protein that can be released by several types of cell. It plays an important part in the immune response.

This protein binds to specific receptors on the surface of cell membranes stimulating chemical reactions inside the cell. As a result, a number of responses may occur that include inflammation and the stimulation of phagocytosis by macrophages.

Antibodies to TNF are used to treat rheumatoid arthritis.

Scientists have used a virus that infects bacteria to produce antibodies to human TNF.

The diagram summarises this process.



- (i) Explain how attaching the gene for TNF to the viral gene results in TNF being displayed on the outside of the capsid.

(2)

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(ii) Explain why antibodies to TNF are used in the treatment of rheumatoid arthritis.

(2)

(iii) Patients being treated with antibodies to TNF are more susceptible to tuberculosis (TB), which can be fatal.

Explain why patients being treated with antibodies to TNF can die from TB.

(4)

(Total for Question 4 = 10 marks)